Study guide: Paper 010

AuthorityEffect: None. Study first, attempt the quiz, then use the separate instructor key. Definitions come from manuscript Section 2; pilot data and limits come from Sections 3–6.

1. Validate the algebra

A finite table must be total and closed, have a two-sided identity and inverses, and be associative for every triple. Names and values need their declared types. The S3 construction uses (p q)(x)=p(q(x)); Q8 uses integer quaternions. No floating-point tolerance enters these checks.

In Q8, i j (−i) (−j)=−1, while i(−i)j(−j)=1. The code labels 1 and −1 as `e` and `-e`. A −1 square is non-flat even though −1 is central and self-inverse.

2. Keep the path convention visible

Plaquettes are (r,c), (r+1,c), (r+1,c+1), (r,c+1). The last two edges use inverses of stored positive links. Reversal inverts the product. Moving the basepoint to the next vertex conjugates by the inverse of the first edge. Adjacent gauge factors cancel along a path, leaving `g(start) hol(path) inverse(g(end))`. A closed product is conjugated at its basepoint: identity status persists, but its exact element can change.

Use a non-central S3 product of order three to detect reversal errors. The Q8 commutator −1 alone cannot distinguish its two orientations.

3. Local flatness and global gauge triviality

The declared lattice includes its plaquettes. An open rectangle has trivial fundamental group, so flatness lets a rooted spanning-tree gauge trivialize every edge. On a torus, flatness leaves commuting cycle products A and B; both must be identity for a trivializing gauge. A=−1, B=1 therefore gives `FLAT_NONTRIVIAL`.

The host accepts open dimensions at least 2 and torus dimensions at least 3. Endpoint-only names cannot distinguish parallel torus edges at size 2. For seam links a,b, only the seam corner can be non-flat, with product b a b ^{1} a ^{1}. Enumerating $S3^{2}$ and $Q8^{2}$ covers 36+64=100 connections in this family, not all connections.

4. Optimization and proof boundary

Four identity links have identity product. Skipping their multiplications preserves the complete check; every other plaquette is still tested. The number skipped is gauge-dependent. Group validation takes $O(|G|^{3})$; after validation, the lattice pass and tree work take O(LM). A submitted path of k edges adds O(k) work.

Lean formalizes only the four-identity-link observation. The named report lists one axiom-free theorem. The historical full-module report lists 26 declarations: 25 axiom-free and one using `propext`, including generated structure declarations. Neither report formalizes the parser, flatness host or pilot analysis. Read the current receipt to distinguish static report checks from an executed build.

5. Require a witness

Verdict	Checked obligation	Near miss
NOT_FLAT	Actual plaquette with its matching nonidentity product	Correct label, wrong product or winding four-cycle
FLAT_TRIVIAL / PURE_GAUGE	Full gauge making every positive link identity	Missing vertex, conflicting alias or a remaining nonidentity edge
FLAT_NONTRIVIAL	All plaquettes flat and a closed path with matching nonidentity product	Open path, identity loop or non-flat input

The verifier accepts valid plaquette rotations and either orientation. Parsed edge or vertex aliases cannot overwrite entries. Unknown elements are refused before a verdict rather than treated as identity or silently omitted.

6. The right C017 denominator

Twenty-five historical responses are recorded. Fifteen solution records are retained: 13 parsed certificates and two parser-error wrappers. Fresh host checks accept 10 and reject 5 overall, including both wrappers. The other ten retain labels/reasons and zero recorded acceptances; they cannot be freshly certificate-checked. All 25 extracted entries can be compared with the original reports, which is a different check.

Arm 1's labels match 4/5 while certificates score 0/5 under the old harness. Three refusals concern vertex order before products were compared; those products are missing. The current host accepts either orientation but cannot recover absent payloads.

7. Audit historical tables

Table 3 reports original checker decisions out of ten per cell. C049 has two different panels; C050 repeats one panel. Table 4 uses the author's case-by-case classification (36 of 60 identifiers name a verdict/answer) and a prior checker

replay. Recounting does not independently assign these categories or measure model reliance on leakage. Readers can dispute the published annotations.

Always refusing passes 10/10 in original C045 and C046. Acceptance there cannot separate mathematical answers from that baseline. Paired checkers may share code or permissive failure modes; inspect obligations and negative controls before treating agreement as added assurance.

8. Name the execution supporting the claim

Operation	Evidence produced	Limit
C017 replay	Fresh verdicts and 15 solution-record checks	Ten payloads remain missing; no new responses
Pilot recount	Hashes, 25 extraction comparisons, Table 3/4 counts	Original flags and prior record; no fresh C045–C050 checker execution
Lean build/census	Result under the actual pinned compiler	Only the identity lemma; confirm the build ran
Deposit verification	Membership and byte custody	Does not establish mathematical truth
Hosted-model trial	New responses under recorded settings	Requires actual requests and suitable experimental design

Run the capstone and explain its fields. Future performance work needs opaque IDs, stripped answer notes, repaired checker refusal rules, negative controls, matched conditions/budgets, provider/error metadata, predeclared samples and complete retained responses. The package provides no new provider trial, learner study, physical result or operational authority.